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**Assignment: 01**

**Subject: Deep Learning**

## **Assignment 1 – Solving XOR with a Deep Neural Network**

This assignment demonstrates how to solve the classic XOR problem using a Deep Neural Network (DNN) built from scratch in NumPy. The XOR function is not linearly separable and requires non-linear modeling to be effectively represented.

### **Architecture:**

- Input Layer: 2 units (binary inputs).
- Hidden Layer: 4 neurons with tanh activation.
- Output Layer: 1 neuron with sigmoid activation.
- Loss Function: Binary Cross-Entropy.
- Optimizer: Gradient Descent (from scratch).
- Learning Rate: 0.1, Epochs: up to 20,000 with early stopping.

### **Results:**

- Training converged in 59 epochs.
- Final Loss: 0.603267.
- Predictions matched expected XOR outputs:
  - (0,0) → 0
  - (0,1) → 1
  - (1,0) → 1
  - (1,1) → 0

**Why XOR Needs Non-Linearity:**

The XOR dataset cannot be separated by a single linear boundary because positive class points (0,1) and (1,0) lie diagonally. A hidden layer with non-linear activation (e.g., tanh) enables the network to combine multiple linear regions into a non-linear decision boundary, solving XOR.

